

Original Investigation | CLINICAL SCIENCES

Age, Survival Predictors, and Metastatic Death in Patients With Choroidal Melanoma

Tentative Evidence of a Therapeutic Effect on Survival

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← Editorial page 519

IMPORTANCE The influence of ocular treatment of choroidal melanoma on survival has yet to be elucidated.

OBJECTIVE To determine whether treatment of choroidal melanoma influences survival by correlating age at death, cause of death, age at treatment, and survival predictors.

DESIGN, SETTING, AND PARTICIPANTS Prospective cohort study performed at the Liverpool Ocular Oncology Centre, a supraregional, tertiary referral service in England. We included 3072 patients treated for choroidal melanoma from January 15, 1993, through November 23, 2012, and who reside in the mainland United Kingdom.

EXPOSURES A diagnosis of choroidal melanoma (ie, any uveal melanoma involving the choroid).

MAIN OUTCOMES AND MEASURES Largest basal tumor diameter, tumor thickness, TNM stage, ciliary body involvement, extraocular spread, melanoma cytomorphological findings, closed connective tissue loops, mitotic count, chromosome 3 loss, chromosome 6p gain, chromosome 8q gain, age at treatment, age at death, and cause of death.

RESULTS The largest basal tumor diameter correlated with all survival predictors except for chromosome 6p gain. Older age at treatment correlated with ciliary body involvement, extraocular spread, largest basal tumor diameter, tumor thickness, TNM stage, epithelioid cells, chromosome 3 loss, and chromosome 8q gain. A total of 1005 patients had died by the close of the study. The cause of death was metastatic disease due to uveal melanoma in 561 patients. Among the 561 patients, survival time after treatment correlated with sex, basal tumor diameter, ciliary body involvement, extraocular spread, TNM stage, closed loops, and mitotic count. In this group of patients, none of the survival predictors correlated with age at death except for mitotic count, which showed a weak correlation. All survival predictors correlated with an increased likelihood of metastatic melanoma as the cause of death.

CONCLUSIONS AND RELEVANCE Patients who are younger at the time of treatment tend to have a smaller, less extensive tumor with a lower degree of malignancy. A tentative explanation for these findings is that ocular treatment prevents tumor growth, dedifferentiation, and metastatic disease in some patients, especially those with a smaller tumor.

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Almost 50% of all patients with choroidal melanoma die of metastatic disease, despite successful eradication of the primary ocular tumor.¹ In 1978, Zimmerman et al² suggested that enucleation might accelerate death due to metastatic disease. This hypothesis encouraged the Collaborative Ocular Melanoma Study Group^{3,4} to compare enucleation with (1) iodine plaque brachytherapy and (2) enucleation combined with neoadjuvant radiotherapy in a series of randomized studies. These studies concluded that the prevention of metastatic spread was achieved as effectively with enucleation as with both comparison treatments. In fact, these study findings were inconclusive because many patients already had metastatic spread by the time of diagnosis and treatment, as evidenced by the short survival times.⁵ Because prevention of what has already happened is impossible, such patients should have been excluded from the studies, which would have left insufficient numbers of patients to achieve adequate statistical power. Therefore, whether ocular treatment influences survival, and if so in whom, remains to be shown.

In theory, the best way to investigate how ocular treatment affects survival would be to perform a randomized study comparing immediate treatment with nontreatment. In practice, such a study would be difficult if not impossible to perform because of ethical concerns about leaving cancer untreated and because many patients would drop out of the study if tumor growth is observed or if they develop symptoms.

Many patients with uveal melanoma experience a delay in the treatment of their tumor, because they present late in the disease course or because their tumor is missed when they undergo ophthalmoscopy.⁶ By comparing survival of such patients with survival of patients who undergo early treatment, insights into the vital effect of ocular treatment are possible. Studies have consistently demonstrated that survival time after treatment of large tumors is shorter than that after treatment of small melanomas.⁷ We do not know whether this observation indicates a therapeutic effect or whether it merely reflects lead-time bias, with larger tumors having been growing and metastasizing for a longer time.

In this study, we sought to overcome the problem of lead-time bias by measuring total life span instead of survival time after treatment. We hypothesized that if early ocular treatment of uveal melanoma indeed prolongs life, then patients with a smaller tumor at the time of treatment will tend to live longer and be more likely to die of unrelated disease. If therapy is inconsequential, then the life span and cause of death of all patients will be the same irrespective of tumor size and the degree of malignancy. Tumor size may not only reflect therapeutic delay; it may also indicate tumor growth rate and degree of malignancy so that the shorter life spans of patients with large tumors might reflect only their melanoma-doubling time. To address this question, we also correlated the tumor size with age at presentation. We hypothesized that if large tumor size indicates delayed treatment, then patients with a large tumor should be older, whereas if large tumor size reflects a faster tumor growth rate, then patients with a large tumor should be younger at the time of tumor detection and treatment.

To perform this study, we made 2 assumptions. First, because the age at presentation with uveal melanoma shows a

sharp peak around 60 years, we assumed that a similar peak age occurs at which these tumors first arise. One can only speculate when this peak age might be, but in any case the answer is immaterial to the present study. Second, we assumed that the degree of malignancy of posterior uveal melanomas is not influenced by the patient's age when the tumor first develops, so that the rate of tumor growth is not influenced by the patient's age. In summary, the aim of this study was to determine whether the life span of patients with posterior uveal melanoma is prolonged and the cause of death is not metastatic disease when the tumor undergoes early ocular treatment.

Methods

Inclusion Criteria

Patients were included in this study if (1) they were diagnosed as having choroidal melanoma (ie, any uveal melanoma involving the choroid) at the Liverpool Ocular Oncology Centre from January 15, 1993, through November 23, 2012; (2) the largest basal tumor diameter at the time of treatment was known; (3) the date of treatment was recorded (some patients declined treatment or were treated elsewhere); and (4) the patient resided in the mainland United Kingdom (ie, England, Scotland, or Wales). This study adhered to the tenets of the Declaration of Helsinki. Prospective written consent for the use of data, tissues and images for research, teaching, and audit was routinely obtained from patients. Institutional review board approval was not required.

Identification of Patients in Database

The study cohort was identified by searching the computerized database of the Liverpool Ocular Oncology Centre. The database contained information that had been entered routinely and contemporaneously by a full-time data manager.

Clinical Examination and Diagnostic Methods

At their initial assessment at our center, all patients routinely underwent full ophthalmic examination with slitlamp biomicroscopy, gonioscopy if indicated, and indirect ophthalmoscopy. All patients were also examined by fundus photography, B-scan ultrasonography, and in selected cases, autofluorescence, fluorescein angiography, indocyanine green angiography, and/or optical coherence tomography. The diagnosis of uveal melanoma was based on clinical features (eg, visual symptoms, tumor size, lipofuscin pigment, retinal detachment) and ultrasonographic appearance (eg, thickness >2 mm, collar-stud shape). In a few patients, the diagnosis was established by results of transretinal biopsy.

Tumor dimensions were obtained by B-scan ultrasonography using a variety of machines. The tumor thickness was measured from the internal scleral surface, with the direction of gaze being toward the quadrant of the tumor if this lesion was located peripherally. The ultrasonography was facilitated by the examiner performing binocular indirect ophthalmoscopy immediately before so that special care could be taken to define tumor margins if these were tapering.

Treatment Methods

Patients were treated by radiotherapy (with ruthenium plaque or proton beam radiotherapy), surgical excision (ie, exoresection, endoresection, or enucleation), and/or phototherapy (ie, transpupillary thermotherapy or photodynamic therapy). Treatment was selected according to tumor size, location, and extent, taking into account the patient's needs and wishes.

Histological Examination

Histological examination was performed on all eyes that underwent enucleation or local resection and on all tumors that underwent biopsy. Until 2002, tumor specimens were routinely fixed in glutaraldehyde. After that date, buffered formalin was used. Histological examination was performed using sections stained with hematoxylin-eosin and, if necessary, immunohistochemical analysis using melan A. Melanomas were categorized as spindle-cell, epithelioid, or mixed type. They were recorded as having epithelioid cells irrespective of the proportion of such cells in the tumor. From 1994 onward, extravascular matrix patterns were assessed to identify closed connective tissue loops using the periodic acid-Schiff reagent, without counterstaining. Mitoses were counted in 40 high-power fields (objective $\times 40$) in hematoxylin-eosin-stained sections. Extraocular extension was recorded as being present whether noted clinically or on pathological examination.

Genetic Studies

Tumors underwent analysis for chromosome 3 loss using microsatellite analysis during 1999, 2000, and after 2010; fluorescence in situ hybridization from 1999 to 2007; and multiplex ligation-dependent probe amplification from 2006 onward, with some overlap of techniques during transition periods.⁸ These tests were routinely performed on fresh tumor samples. For the purposes of this study, only multiplex ligation-dependent probe amplification data were used because this method was the most informative and reliable.

Follow-up Methods

After ocular treatment, patients underwent review after 1 to 4 weeks, then every 6 months for 3 to 5 years, and then once a year indefinitely. Follow-up assessments were usually alternated between our center and the referring hospital until the risk of local tumor recurrence was considered to be small (ie, <1%), when the patient was discharged from our unit to the referring ophthalmologist.

We notified the Cancer Registry of the National Health Service of all patients, who were flagged by their National Health Service number. The Cancer Registry automatically informed us of the date and cause of death of each patient, usually within a few weeks of this event.

Statistical Analysis

We analyzed the data using commercially available statistical software (SPSS; SPSS Inc). We used the χ^2 test to analyze categorical variables; the Kruskal-Wallis, Mann-Whitney, or Spearman test for continuous variables; and the log-rank test for survival.

Results

Age and Tumor Features at Treatment

The 3072 patients included 1579 males (51.4%) and 1493 females (48.6%) (Table 1). The tumors had a median basal diameter of 12.1 (range, 2.4-23.8) mm and a median thickness of 4.1 (range, 0.6-18.3) mm. The tumor cell type was known in 1624 patients (52.9%) and the chromosome 3 status was known in 570 (18.6%).

The largest basal tumor diameter correlated with all survival predictors except for chromosome 6p gain (Table 2). The primary treatment consisted of brachytherapy (1130 [36.8%]), enucleation (954 [31.1%]), proton beam radiotherapy (643 [20.9%]), transscleral resection (197 [6.4%]), endoresection (95 [3.1%]), and phototherapy (53 [1.7%]).

The median age at treatment was 62.8 (range, 12.4-96.9) years. Older age at treatment correlated with ciliary body involvement, extraocular spread, largest basal tumor diameter, tumor thickness, TNM stage, chromosome 3 loss, and chromosome 8q gain but not with mitotic count and the presence of closed loops (Table 1). The median age at treatment was 60.7 years in patients with a basal tumor diameter less than 10 mm and 65.6 years if the basal tumor diameter exceeded 18 mm, a difference of 4.9 years (Table 1).

Life Span and Cause of Death

A total of 1005 patients had died by the close of the study (Table 3). The cause of death was metastatic disease from uveal melanoma in 561 patients. The largest basal tumor diameter of these patients had a median of 15.4 (range, 5.3-23.8) mm. The median age at death due to any cause was 74.0 years, whereas the median age at death due to metastatic melanoma was 68.6 years.

In patients who died of metastatic disease, the survival time after treatment correlated with sex, basal tumor diameter, ciliary body involvement, extraocular spread, TNM stage, closed loops, and mitotic count (Table 4). Among these patients who died of metastasis, none of the survival predictors correlated with age at death except for mitotic count, which showed a weak correlation (Spearman $P = .01$) (Table 4).

Table 3 also shows the proportion of patients who died of metastatic disease among all causes of death according to the survival predictors investigated. All survival predictors correlated with an increased likelihood of the cause of death being metastatic melanoma.

Discussion

Main Findings

In patients who were older when their choroidal melanoma was treated, the tumor tended to be larger and more extensive, to have a greater degree of histological malignancy, and to be more likely to show chromosome 3 loss and, hence, metastatic potential. Tumor size and extent also correlated directly with histological and genetic survival predictors. Patients who died, as a group, had a shorter median life span

if risk factors for metastasis were present (ie, if the tumor was more advanced with a greater degree of malignancy). In patients dying of metastatic disease from their uveal melanoma, the life span did not correlate significantly with any of the predictive factors for metastasis (except for a weak correlation with mitotic count). The time from ocular treatment to

death due to metastasis was shorter in patients with predictive factors for metastasis (ie, if the tumor was more advanced with a greater degree of malignancy). Among the patients who died, the likelihood of death being caused by metastasis was higher in those whose tumor showed clinical, histological, and/or genetic risk factors for metastasis.

Table 1. Correlation Between Age at Treatment and Risk Factors for Metastasis^a

Variable	No. (%) of Patients	Median Age at Treatment, y	P Value (Test)
Sex			
Male	1579 (51.4)	62.8	.38 (Mann-Whitney)
Female	1493 (48.6)	62.8	
Basal tumor diameter, mm			
<10	690 (22.5)	60.7	<.001 (Spearman)
10-12	954 (31.1)	62.8	
13-15	750 (24.4)	62.7	
16-18	457 (14.9)	64.4	
>18	221 (7.2)	65.6	
Tumor thickness, mm			
<4	1294 (42.1)	62.6	.03 (Spearman)
4-6	840 (27.3)	62.6	
7-9	531 (17.3)	62.2	
>9	407 (13.2)	64.6	
Ciliary body involvement			
No	2305 (75.0)	62.3	<.001 (Mann-Whitney)
Yes	767 (25.0)	64.1	
Extraocular spread			
No	2872 (93.5)	62.6	<.001 (Mann-Whitney)
Yes	200 (6.5)	67.1	
TNM stage			
I	925 (30.1)	61.6	<.001 (Kruskal-Wallis)
II	1448 (47.1)	62.8	
III	699 (22.8)	64.4	
Cytomorphological findings			
Spindle	624 (38.4)	61.4	.004 (Mann-Whitney)
Epithelioid	1000 (61.6)	63.5	
Closed loops			
No	549 (49.6)	63.1	.27 (Mann-Whitney)
Yes	558 (50.4)	64.4	
Mitotic count by HPF (objective ×40)			
≤4.4	705 (60.8)	63.3	.85 (Spearman)
≥4.5	454 (39.2)	64.0	
Chromosome 3 loss			
Nil	189 (33.2)	60.3	<.001 (Kruskal-Wallis)
1-7 Loci	117 (20.5)	61.4	
>7 Loci	264 (46.3)	64.6	
Chromosome 6p gain			
Nil	424 (74.4)	63.1	.053 (Kruskal-Wallis)
Partial	114 (20.0)	61.6	
Total	32 (5.6)	60.1	
Chromosome 8q gain			
Nil	295 (51.8)	62.3	.04 (Kruskal-Wallis)
Partial	217 (38.1)	62.2	
Total	58 (10.2)	67.8	

Abbreviation: HPF, high-power field.

^a Includes 3072 patients. Data were not available for all variables in all patients. Percentages have been rounded and may not total 100.

Inferences From Main Findings

If the development of choroidal melanomas does indeed cluster at a peak age, in a similar manner to the clustering of age at presentation and age at death, then being older at treatment could indicate a longer delay in tumor diagnosis and treatment. Such delay would be expected to allow tumors to become larger and more extensive and have a greater degree of malignancy, with some developing metastatic potential and dissemination as a result of the delay. For example, if tumors with a basal diameter of less than 10 mm present at a

median age of 60 years, and if tumors with a basal diameter exceeding 18 mm present at a median age of 65 years, this finding would suggest that, on average, it takes 5 years for a tumor to grow from less than 10 mm to more than 18 mm in basal diameter. If the prevalence of chromosome 3 loss is 54% in tumors smaller than 10 mm in basal diameter and 90% in tumors with a basal diameter exceeding 18 mm, one might expect that tumor growth from 10 to 18 mm is associated with the development of chromosome 3 loss in a number of patients. Our findings tentatively support these hypotheses,

Table 2. Correlation Between Largest Basal Tumor Diameter and Risk Factors for Metastasis^a

		Largest Basal Tumor Diameter, % of Patients					
Variable	No. (%) of Patients	<10 mm	10-12 mm	13-15 mm	16-18 mm	>18 mm	P Value (Test)
Sex							
Male	1579 (51.4)	48.8	50.5	48.8	56.5	61.5	.001 (Mann-Whitney)
Female	1493 (48.6)	51.2	49.5	51.2	43.5	38.5	
Tumor thickness, mm							
<4	1294 (42.1)	87.2	51.4	19.6	9.8	4.5	<.001 (Spearman)
4-6	840 (27.3)	10.3	32.1	44.3	20.1	17.6	
7-9	531 (17.3)	1.9	12.4	22.9	36.8	27.1	
>9	407 (13.2)	0.6	4.2	13.2	33.3	50.7	
Ciliary body involvement							
No	2305 (75.0)	93.3	86.7	72.9	49.0	28.5	<.001 (Mann-Whitney)
Yes	767 (25.0)	6.7	13.3	27.1	51.0	71.5	
Extraocular spread							
No	2872 (93.5)	98.1	97.9	93.1	84.2	80.5	<.001 (Mann-Whitney)
Yes	200 (6.5)	1.9	2.1	6.9	15.8	19.5	
TNM stage							
I	925 (30.1)	87.5	33.6	0	0	0	<.001 (Kruskal-Wallis)
II	1448 (47.1)	12.2	62.6	78.3	39.4	0	
III	699 (22.8)	0.3	3.8	21.7	60.6	100.0	
Cytomorphological findings							
Spindle	624 (38.4)	54.5	44.0	39.5	29.2	28.2	<.001 (Mann-Whitney)
Epithelioid	1000 (61.6)	45.5	56.0	60.5	70.8	71.8	
Closed loops							
No	549 (49.6)	74.6	65.3	49.7	41.1	38.2	<.001 (Mann-Whitney)
Yes	558 (50.4)	25.4	34.7	50.3	58.9	61.8	
Mitotic count by HPF (objective ×40)							
≤4.4	705 (60.8)	87.5	77.6	65.1	50.4	42.7	<.001 (Spearman)
≥4.5	454 (39.2)	12.5	22.4	34.9	49.6	57.3	
Chromosome 3 loss							
Nil	189 (33.2)	46.4	46.0	43.6	17.3	10.5	<.001 (Kruskal-Wallis)
1-7 Loci	117 (20.5)	25.0	23.7	18.1	20.7	15.8	
>7 Loci	264 (46.3)	28.6	30.2	38.3	62.0	73.7	
Chromosome 6p gain							
Nil	424 (74.4)	91.1	75.5	67.1	73.3	76.3	.856 (Kruskal-Wallis)
Partial	114 (20.0)	7.1	20.1	24.8	20.0	19.7	
Total	32 (5.6)	1.8	4.3	8.1	6.7	3.9	
Chromosome 8q gain							
Nil	295 (51.8)	69.6	65.5	53.7	43.3	26.3	<.001 (Kruskal-Wallis)
Partial	217 (38.1)	28.6	29.5	38.3	46.7	43.4	
Total	58 (10.2)	1.8	5.0	8.1	10.0	30.3	

Abbreviation: HPF, high-power field.

^a Includes 3072 patients. Data were not available for all variables in all patients. Percentages have been rounded and may not total 100.

suggesting that early treatment should prevent not only tumor growth and local invasion, but also dedifferentiation and metastatic spread in some patients. Closed loops and mitotic count did not correlate with age at treatment. Explanations for these results could be lack of change with age,

inaccurate measurement, and smaller numbers of patients, because these features cannot be assessed in biopsy specimens. Chromosome 6p gain, which correlates with good prognosis, showed an inverse correlation with age, possibly because of intratumoral cellular heterogeneity, with such cells

Table 3. Correlations Between Risk Factors for Metastasis and Age at and Cause of Death^a

Variable	No. (%) of Patients	Median Age at Death, y	P Value (Test)	Deaths Due to Metastatic Disease, % ^b	P Value (Test)
Sex					
Male	537 (53.4)	72.6	<.001 (Mann-Whitney)	59.2	.02 (χ ²)
Female	468 (46.6)	76.0		51.9	
Basal tumor diameter, mm					
<10	127 (12.6)	77.7	<.001 (Spearman)	15.0	<.001 (Kruskal-Wallis)
10-12	235 (23.4)	77.4		37.4	
13-15	281 (28.0)	72.4		64.1	
16-18	224 (22.3)	72.0		72.3	
>18	138 (13.7)	69.8		81.2	
Tumor thickness, mm					
<4	246 (24.5)	77.5	<.001 (Spearman)	28.0	<.001 (Kruskal-Wallis)
4-6	306 (30.4)	74.0		54.2	
7-9	239 (23.8)	70.7		67.8	
>9	214 (21.3)	71.5		76.6	
Ciliary body involvement					
No	628 (62.5)	75.5	<.001 (Mann-Whitney)	45.5	<.001 (χ ²)
Yes	377 (37.5)	70.9		72.9	
Extraocular spread					
No	881 (87.7)	74.1	.422 (Mann-Whitney)	52.6	<.001 (χ ²)
Yes	124 (12.3)	72.9		79.0	
TNM stage					
I	163 (16.2)	77.7	<.001 (Kruskal-Wallis)	17.2	<.001 (χ ²)
II	443 (44.1)	75.4		50.8	
III	399 (39.7)	70.1		77.2	
Cytomorphological findings					
Spindle	136 (22.1)	73.7	.03 (Mann-Whitney)	49.3	<.001 (χ ²)
Epithelioid	479 (77.9)	70.9		74.3	
Closed loops					
No	189 (36.6)	73.5	.003 (Mann-Whitney)	58.2	<.001 (χ ²)
Yes	327 (63.4)	70.3		79.8	
Mitotic count by HPF (objective ×40)					
≤4.4	290 (52.0)	75.2	<.001 (Spearman)	58.6	<.001 (Kruskal-Wallis)
≥4.5	268 (48.0)	69.3		81.0	
Chromosome 3 loss					
Nil	14 (10.2)	81.3	<.001 (Kruskal-Wallis)	28.6	<.001 (χ ²)
1-7 Loci	19 (13.9)	69.2		73.7	
>7 Loci	104 (75.9)	67.6		90.4	
Chromosome 6p gain					
Nil	116 (84.7)	67.9	.32 (Mann-Whitney)	87.1	.001 (χ ²)
Partial	19 (13.9)	73.5		52.6	
Total	2 (1.5)	76.8		50.0	
Chromosome 8q gain					
Nil	48 (35.0)	68.7	>.99 (Mann-Whitney)	68.8	.02 (χ ²)
Partial	57 (41.6)	70.3		89.5	
Total	32 (23.4)	73.0		87.5	

Abbreviation: HPF, high-power field.

^a Includes 1005 patients who died during the study period. Data were not available for all patients. Percentages have been rounded and may not total 100.

^b Numbers of patients in each category who died of metastatic disease are given in Table 4.

being overgrown by more rapidly proliferating clones of cells not having this feature.

The largest basal tumor diameter correlated significantly with all histological and genetic survival predictors except chromosome 6p gain. The increase in the prevalence of these predictors with age at treatment suggests that they tend to

arise concurrently with tumor growth. An alternative explanation is that uveal melanomas tend to dedifferentiate and to grow more rapidly if they arise at an older age than if they develop in younger individuals. This hypothesis is not supported by our finding that mitotic count, which reflects melanoma doubling time, did not correlate with age. Some scope

Table 4. Correlation Between Time From Treatment to Metastatic Death and Age at Metastatic Death and Risk Factors for Metastasis^a

Variable	No. (%) of Patients	Median Survival, y	P Value ^b	Median Age at Death, y	P Value (Test)
Sex					
Male	318 (56.7)	3.0	.04	68.3	.51 (Mann-Whitney)
Female	243 (43.3)	3.4		68.8	
Basal tumor diameter, mm					
<10	19 (3.4)	4.8	<.001	75.6	.32 (Spearman)
10-12	88 (15.7)	4.4		69.0	
13-15	180 (32.1)	4.1		67.2	
16-18	162 (28.9)	2.6		68.7	
>18	112 (20.0)	2.0		67.8	
Tumor thickness, mm					
<4	69 (12.3)	4.2	.18	71.6	.40 (Spearman)
4-6	166 (29.6)	3.5		66.6	
7-9	162 (28.9)	3.0		67.2	
>9	164 (29.2)	2.9		69.4	
Ciliary body involvement					
No	286 (51.0)	3.9	<.001	69.0	.38 (Mann-Whitney)
Yes	275 (49.0)	2.7		68.2	
Extraocular spread					
No	463 (82.5)	3.4	<.001	67.7	.10 (Mann-Whitney)
Yes	98 (17.5)	2.2		70.0	
TNM stage					
I	28 (5.0)	4.6	<.001	73.6	.35 (Kruskal-Wallis)
II	225 (40.1)	4.0		68.8	
III	308 (54.9)	2.7		68.1	
Cytomorphological findings					
Spindle	67 (15.8)	3.5	.13	68.3	.58 (Mann-Whitney)
Epithelioid	356 (84.2)	2.9		68.3	
Closed loops					
No	110 (29.6)	3.6	<.001	70.3	.15 (Mann-Whitney)
Yes	261 (70.4)	2.6		68.0	
Mitotic count by HPF (objective ×40)					
≤4.4	170 (43.9)	3.5	.03	70.7	.01 (Spearman)
≥4.5	217 (56.1)	2.5		67.1	
Chromosome 3 loss					
Nil	4 (3.6)	3.1	.42	74.8	.44 (Kruskal-Wallis)
1-7 Loci	14 (12.5)	2.5		69.7	
>7 Loci	94 (83.9)	2.5		67.2	
Chromosome 6p gain					
Nil	101 (90.2)	2.6	.56	67.3	.89 (Kruskal-Wallis)
Partial	10 (8.9)	1.8		70.7	
Total	1 (0.9)	5.7		72.3	
Chromosome 8q gain					
Nil	33 (29.5)	2.3	.65	66.3	.70 (Kruskal-Wallis)
Partial	51 (45.5)	2.6		69.1	
Total	28 (25.0)	2.5		68.3	

Abbreviation: HPF, high-power field.

^a Includes 561 patients who died of metastatic disease. Data were not available for all variables in all patients. Percentages have been rounded and may not total 100.

^b Calculated using the log-rank test.

exists for studies investigating histological and clinical features of malignant disease at different ages after adjusting for tumor size.

The median life span of all patients who died of any cause was shorter in the presence of risk factors for metastasis. This probably reflects the rate of metastatic deaths (ie, the percentage of patients dying of metastasis) and not the number of years of life lost, which in an individual patient mostly depends on age at treatment.

The life span of patients dying of metastatic disease did not correlate with any of the survival predictors, except for mitotic count, which showed a weak correlation. The trends we observed suggest that this lack of statistical significance may be due to insufficient patient numbers. It would seem worthwhile repeating this study with a larger cohort to determine whether early treatment prolongs life span in patients dying of metastasis.

The time from ocular treatment to metastatic death correlated significantly with basal tumor diameter and most other survival predictors. These findings probably reflect lead-time bias and more rapid growth of more malignant tumors. Chromosome 3 loss did not correlate statistically with survival time, probably because only 4 patients who died of metastasis did not have this abnormality, which may have been missed by the laboratory technique that was used.

This study found statistically significant correlations between all risk factors for metastasis and the likelihood of the cause of death being metastatic disease. For example, in patients who died, the percentage of deaths due to metastatic disease increased from 15% to 82% with the increase in basal diameter from less than 10 to greater than 18 mm.

Strengths and Weaknesses

The main strengths of this study are (1) the large number of patients; (2) the consistent manner in which tumors were measured and categorized; and (3) the completeness of follow-up data. By measuring the patients' life span rather than the survival time after treatment, lead-time bias has been avoided but replaced by other forms of bias.

The main weakness is that inferences regarding therapeutic effect are based on nonrandomized data, so unknown factors may have influenced the results. As mentioned, randomization of such patients is not possible, for ethical and logistic reasons. Another weakness is that some of the smaller tumors may have been nevi; however, such cases are likely to be rare. We diagnosed melanoma according to widely accepted signs of malignant disease (size, serous retinal detachment and confluent lipofuscin). The only way to overcome this problem would have been to perform a biopsy on all patients, which is not conventional practice. Another weakness, as mentioned, is that many of the patients were still alive at the close of the study, so significant bias may have occurred, making it necessary to interpret median life spans and causes of death with caution. These outcomes are best studied on cohorts in which all patients have died, which could take many decades. Any such cohorts in existence would be valuable, and continuing survival studies indefinitely would also be useful.

Published Literature

Straatsma and colleagues⁹ compared survival time after immediate treatment of uveal melanomas with that after delayed treatment. A trend toward increased mortality after delayed treatment was reported; however, bias may have occurred if the tumors that grew did so because from the outset they were more aggressive and lethal than those that remained unchanged.

A previous report¹⁰ described a patient with choroidal melanoma whose tumor suddenly grew after several years of indolent behavior, developing a collar-stud shape; examination of the enucleated eye showed the tumor to consist of disomy 3 spindle cells in the older, basal region and monosomy 3 epithelioid cells in the recently developed, apical area. The report suggested that dedifferentiation of the tumor occurred while the patient was undergoing observation. After the article was published, the patient developed fatal metastases, which may have been prevented by prompt treatment. In a series of 452 patients,¹¹ chromosomal abnormalities were shown to accumulate with tumor growth, in keeping with genomic instability, which is a feature of malignant neoplasia.

Clinical Implications

If confirmed by further studies, the main implication of these findings would be that all patients with uveal melanoma should be treated as soon as possible in the hope of prolonging life. Treatment should be administered urgently irrespective of the tumor size. Therapeutic delay caused by failure in the detection and/or diagnosis of uveal melanoma may be fatal in some patients. Several studies^{6,12,13} have shown that many choroidal melanomas are missed by the practitioner, even when the patient presents with symptoms.

Observation of melanocytic choroidal tumors of indeterminate malignancy may be associated with some risk of metastasis. Patients should be made aware of such risk, and informed consent for nontreatment must be documented. Biopsy may be a safer option if the tumor shows any malignant features (ie, particularly serous retinal detachment, orange pigment, and/or thickness >2 mm). Immediate treatment may be safer still, but may be less acceptable to patients, particularly if visual loss or other morbidity is likely. After treatment, any morbidity is less likely to be tolerated without sufficient evidence that the tumor was malignant. Finally, failure of local tumor control by conservative therapy may be life threatening and may contribute to the increased mortality that is seen in patients with local recurrence.¹⁴ Although such recurrence may be an indicator of increased tumor malignancy, the recurrent tumor may itself be fatal.

Conclusions

Risk factors for metastasis from choroidal melanoma are more common in older patients, which suggests that ocular treatment is more likely to prevent tumor growth, dedifferentiation, and metastatic death if administered sooner rather than later. Survival after treatment, life span, and cause of death are poor indicators of the influence of treatment especially if not all patients in the cohort have died.

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